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RESEARCH

Randomized controlled trial of an app for cancer pain management

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Abstract

Purpose The primary objective of this investigation was to devise a mobile application for self-management of cancer-related discomfort, with the overarching goal of enhancing patients' overall well-being. Would the utilization of the self-management application result in an amelioration of life quality compared to conventional follow-up procedures?

Methods Modules were meticulously devised with the collaborative expertise of oncology pain specialists employing the Delphi technique. Reliability of the consultation was assessed using Cronbach's α. After developing the app, a prospective randomized controlled study was conducted to evaluate the app's effect on participants' quality of life. The trial group used the app; the control group received a follow-up telephone consultation. Assessments of quality of life were conducted both at baseline and following a 4-week intervention period.

Results After two rounds of Delphi expert consultation, the functional modules of Pain Guardian were determined to include five functional modules, including pain self-measurement (real-time dynamic recording of pain by patients), patient reminders (reminders of outbreaks of pain disposal, medication, and review), uploading of examination reports, online consultation, health education, and other functional modules. Cronbach's α was 0.81. Overall, 96 patients (including esophageal, gastric, colorectal, nasopharyngeal, pulmonary, pancreatic, breast, ovarian, uterine, bone, thoracic, bladder, cervical, soft tissue sarcoma, mediastinal, and lymphoma) with cancer pain were divided into the trial and control groups. There were no significant differences in basic information and quality of life at baseline between groups. After 4 weeks of intervention, quality of life was significantly higher in the trial group than in the control group. Patients' satisfaction with the app was high (93.7%).

Conclusions The primary obstacle encountered in the development of applications for managing cancer-related discomfort lies in the sensitive nature of the subject matter, potentially leading to patient apprehension regarding application usage for pain management. Consequently, meticulous attention to user preferences and anticipations is imperative, necessitating the creation of an application characterized by user-friendliness and medical efficacy.

Trial registration Chinese Clinical Trials Registry ChiCTR1800016066; http://www.chictr.org.cn/showproj.aspx?proj=27153. Date of Registration: 2018–05-09.

Keywords Cancer pain management · App · Delphi, Quality of life

Background

Cancer pain represents the foremost and profoundly incapacitating complication frequently reported by individuals afflicted with cancer [1]. Failure to alleviate cancer pain promptly may precipitate adverse effects on both mental and physical well-being [2], and it is a complication that also entails huge social costs [3, 4]. Consequently, these challenges engender considerable psychological strain upon the families of cancer patients and impose substantial economic burdens upon the healthcare system [5, 6]. Individuals grappling with cancer may derive potential advantages from employing digital applications, herein

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referred to as "apps," for the amelioration of cancer-related pain, particularly those integrated with instant messaging modules, thereby potentially mitigating their pain thresholds [7].

Technological advancements have paved the way for novel avenues that hold promise for transcending the constraints inherent to the conventional medical service paradigm and addressing the disparities in medical resource availability in China, particularly those prevailing between urban centers and underdeveloped rural locales [8].

Numerous pain management applications have been devised [9, 10]. Nevertheless, these applications frequently overlook the reliability and appropriateness of the information they disseminate and commonly lack a foundation rooted in scientific principles. These deficiencies have been ascribed to a dearth of guidance during the developmental phase concerning the utility of applications, insufficient support, or inadequate involvement from healthcare professionals [11]. Patients devoid of adequate guidance face the peril of experiencing a decline in their condition [12]. In a previous investigation aimed at identifying effective design components for an online educational platform catering to patients, individuals expressed a preference for modules that were personalized, interactive, credible, and lucidly presented [13]. In a previous investigation aimed at identifying effective design components for an online educational platform catering to patients, individuals expressed a preference for modules that were personalized, interactive, credible, and lucidly presented. Furthermore, a mere 23 (8.2%) applications incorporated the presence of a healthcare professional, with only a solitary application subjected to scientific scrutiny [10, 14]. A conspicuous absence of dependable and all-encompassing mobile applications for the management of cancer-related pain was evident, characterized by the deficiency of scientific underpinnings, insufficient involvement of healthcare professionals, and the absence of structured self-management initiatives in numerous applications.

The primary objective of this study was to develop a cancer pain self-management application that embodies rigor, authority, scientific integrity, and practical utility, subsequently evaluating its viability in enhancing the quality of life among patients.

Methods

This study encompassed two phases: (1) delineating the app modules via the Delphi method and (2) formulating the app while gauging its impact on patients' quality of life through a randomized controlled trial (RCT). The schematic depiction in Fig. 1 elucidates the methodology.

Phase 1: literature review

The initial phase involved a systematic review to ascertain the advantages of employing an app for monitoring cancer pain and appraising a suite of modules suitable for such monitoring. The review comprised primary peer-reviewed studies of any design utilizing apps for cancer pain management and intervention. We conducted a meta-analysis of the RCTs. The modules for monitoring cancer pain were selected based on their high prevalence globally.

Phase 2: Delphi procedure

Subsequently, an adapted Delphi procedure was employed in the second phase to pinpoint the modules garnering the highest consensus and to assess their usability in cancer pain monitoring. The Delphi technique is a well-recognized method for obtaining expert consensus on a defined problem, and it is particularly suited to investigate communication strategies when differences in conceptions exist and the body of knowledge is developing [15, 16].

Delphi expert selection

Delphi expert consultations engage specialists in cancer pain management, including clinical pharmacists, clinicians, and nurses. Experts were selected based on the following criteria: (1) more than an undergraduate degree, (2) senior professional title, (3) proficiency in cancer pain management, (4) more than 10 years of work experience in relevant fields and a good understanding of the study, and (5) willingness to participate voluntarily in the research with a high level of motivation [17].

Development of consensus

Evaluation indexes Expert positive coefficient, authority level, and the degree of expert opinion coordination were evaluated. The ratio of the number of returned questionnaires to the number of distributed questionnaires is the recovery rate, which was used to indicate the expert positive coefficient. The degree of authority (Cr) was based on the expert's judgment of the content of the consultation (Ca) and their familiarity with the consultation content (Cs), via the formula $Cr = (Ca + Cs)/2$. The value of Ca was based on four dimensions: clinical work experience (0.8 points), theoretical knowledge analysis (0.6 points), reference to foreign knowledge (0.4 points), and reliance on intuitive perception (0.2 points). The expert's Cs was divided into five levels for quantification: very (1.0 point), relatively (0.8 points),

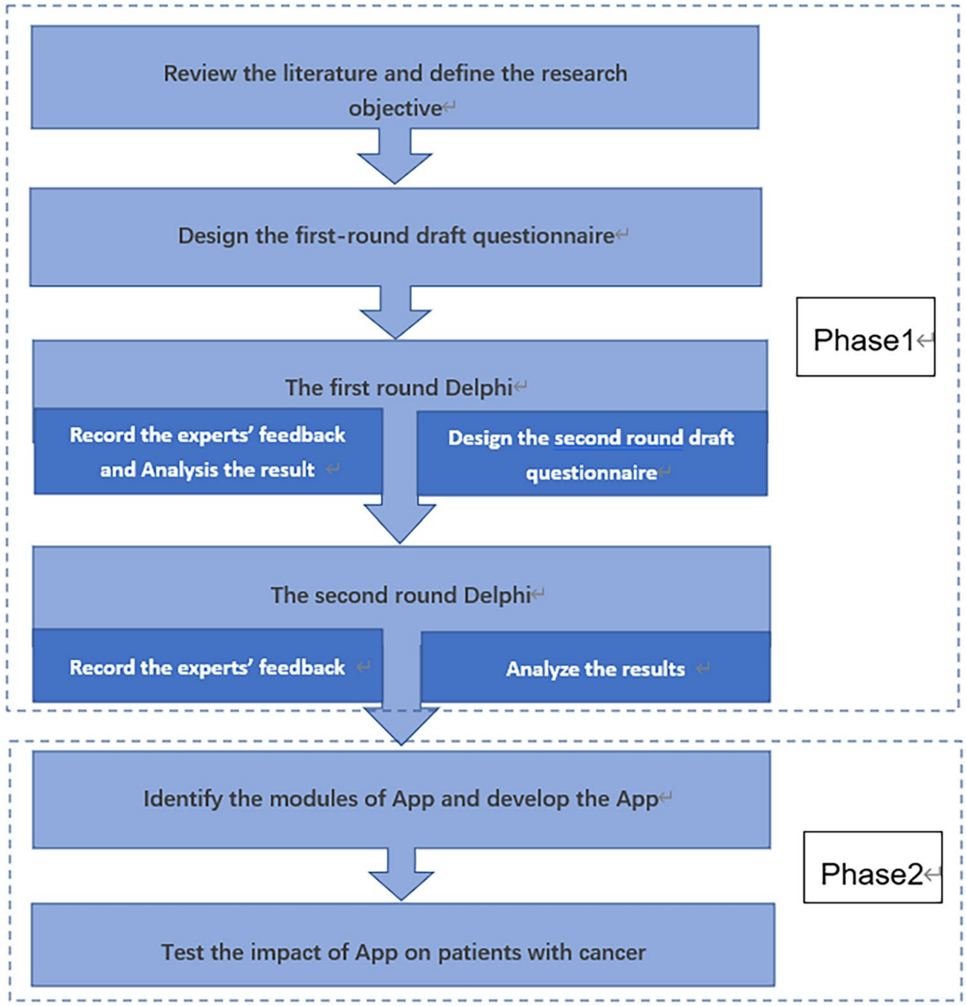

Fig. 1 Methodological workflow

generally (0.6 points), less familiar (0.4 points), and very unfamiliar (0.2 points). [18]

The degree of concordance of expert opinions was expressed by the coefficient of variation (CV). The degree of concentration of expert opinions was expressed by the mean of each module's evaluation. We used Cronbach's α to determine the overall internal consistency of the Cancer Pain Self-management App Modules Questionnaire, which was used to collect the experts' opinions.

Delphi round 1 Following the literature review, we conducted the first round of consultation. Experts were sent an email explaining the study and with a link to a questionnaire, which contained modules that were identified in the review as important to consider when assessing app usability. The experts were asked to rate the modules of the app using a five-point Likert scale (1 = very unimportant and 5 = very important). They were also asked to provide justifications for their ratings.

Between the two rounds Based on the results of the first round of the Delphi questionnaire survey, we conducted the second round of the questionnaire survey. The module selection was based on the following criteria: (1) module importance assignment average (Mj) > 4.5, (2) module full score (Kj, the percentage of experts exhibiting perfect agreement) > 60%, and (3) CV < 0.25. [17]

Delphi round 2 Modules were adjusted based on feedback from the initial round. All experts from Round 1 were invited to partake in Round 2 to reassess modules meeting Round 1 criteria.

Phase 3: testing the app

A study was designed to test the effectiveness of the app for patients with cancer pain. Elaborate on the randomization process for assigning patients to the trial and control groups. The patients in the trial group were trained on how to correctly operate the app. Both groups received an assessment of quality of life at study onset. The core scale utilized for evaluating the quality of life in cancer patients, as part of the system of quality of life scales developed by the European Organization for Research and Treatment of Cancer (EORTC), comprises 30 items. These include 5 functional scales (somatic functioning, role functioning, cognitive functioning, emotional functioning, and social functioning), 3 symptom scales (fatigue, pain, and nausea and vomiting), 6 single-item measures (dysphagia, loss of appetite, sleep disturbances, constipation, diarrhea, and financial difficulties), and 1 patient self-report item (overall health). This comprehensive scale has been widely adopted in clinical trials for its ability to provide a thorough assessment of patients' quality of life. At the end of the trial, the pain and quality of life assessments were repeated in both groups.

Medication compliance of the patients with cancer was evaluated by calculating the ratio of actual consumption of analgesic drugs to theoretical consumption [19].

A feedback item was devised to appraise user satisfaction with the app. Participants completed a questionnaire at the study's conclusion. A five-point Likert scale was used to evaluate satisfaction with the app (5 = very satisfied, 4 = satisfied, 3 = neutral, 2 = dissatisfied, and 1 = very dissatisfied).

Data analysis

Quantitative data were analyzed using IBM SPSS Statistics for Windows, Version 22.0, and Microsoft Excel 2017. We obtained descriptive statistics for each module, including the mean agreement scores and standard deviation. To compare patient data between groups, Shapiro–Wilk test was used to test the normality of data; independent-samples t-test was used to analyze normally distributed data and non-parametric Mann–Whitney U test was used for non-normally distributed data. Chi-square tests were used to test the categorical variables. If categorical variables failed to meet the requirements for a chi-square test, Fisher's exact probability test was used. Significance for all statistical tests was set at $P < 0.05$ with two-tailed analysis and 95% confidence intervals.

Ethical considerations

Ethical approval to conduct this study was obtained from the Medical Ethics Committee of Fujian Medical University Union Hospital. An informed consent statement, including a brief description of the study and the conditions of participation (voluntary participation, confidentiality, and data privacy), was provided for all participants to sign prior to starting the Delphi procedure and clinical research.

Table 1 Demographic information of participants in the Delphi method ($N = 23$)

Characteristics	Values
Age (years)	
Median (interquartile range)	43 (38.5–45.5)
Range	35–53
Sex, *n* (%)	
Female	16 (69.6)
Male	7 (30.4)
Years working (%)	
5	0
10	9 (39.1)
20	11 (47.8)
30	3 (13.0)
Type of expert, *n* (%)	
Nursing	9 (39.1)
Pharmaceutical care	9 (39.1)
Clinical	5 (21.7)
Education, *n* (%)	
Doctoral degree	7 (30.4)
Master's degree	7 (30.4)
Bachelor's degree	9 (39.1)

Results

Determination of modules through literature

A total of 8640 articles were retrieved through systematic literature search. After perusing 40 eligible full-text articles, 13 were reviewed: five RCTs, four before-after studies, two single-arm trials, one prospective cohort study, and one prospective descriptive study [8]. Seven modules emerged from the review: an examination report upload function, reminders, Skype, soothing music, pain self-management, cancer pain science dissemination, and online consultation. These modules were confirmed to be related to improving cancer pain relief. Among them, the apps generally encompassed the functions of telephone follow-up, namely consultation and provision of reminders (i.e., patients were reminded to take their medicine and their progress was reviewed). It was determined that expert evaluations should be conducted for these six modules.

Results of Delphi's expert correspondence

Participant demographic characteristics of Delphi experts

A total of 23 experts consented to participate in this study. They were based in 11 provinces of China: Sichuan, Yunnan, Heilongjiang, Hainan, Hubei, Shanxi, Anhui, Guizhou, Guangdong, Shandong, and Fujian.

Table 1 lists the relevant demographic information of the participants, all of whom participated (100% response rate) in Rounds 1 and 2. Our study used the positive and authority coefficients of experts to measure the reliability of the inquiry. The two rounds of Delphi recorded positive coefficients of 100%. Combining the familiarity of experts and the basis of judgment in the study, the authority coefficient was ≥ 0.7, indicating that the expert authority met the criteria [17].

Round 1

Based on the literature review, we proposed the following six app functions: pain self-management, online consultation, reminder, examination report upload function, soothing music, and cancer pain science dissemination.

Round 2

Based on the Delphi results of the first round of expert consultation, the modules were re-scored, excluding the function of "soothing music." The experts were also asked to provide descriptions for modules that they indicated as inappropriate or irrelevant in an "opinion" column. Based on the expert feedback, we deleted the "soothing music" function. The Cronbach's α coefficient was calculated to be 0.81. Table 2 presents the results of both rounds.

The results of the Delphi method offered a structured approach to obtain expert opinions and consensus on the crucial modules of the app, which encompassed pain self-management, online consultation, reminders, report uploading capability, and the dissemination of cancer pain science. This ensured that the app was developed with the input of experts and adhered to scientific rigor.

Results of clinical studies

User statistics

Ninety-six patients were recruited by physicians during hospital visits to the center. The patients met the following screening criteria: (1) able to read and use mobile phones; (2) aged between 18 and 75 years old; (3) etiology of cancer pain conformed to National Comprehensive Cancer Network Clinical (NCCN) Practice Guidelines on Adult Cancer Pain (2019.V3; i.e., pain directly caused by tumors, including by primary and/or metastatic tumors or tumor treatment); (4) adopted the NCCN adult cancer pain guidelines; and (5) understood the research process, agreed to participate in this clinical study, and signed the informed consent form. Exclusion criteria included (1) severe cognitive impairment; (2) severe hepatic insufficiency (ALT ≥ 2.5 × ULN, AST ≥ 2.5 × ULN, TBIL ≥ 1.5 × ULN) and severe renal insufficiency (Cr ≥ 2.5 × ULN); (3) quality of life assessment could not be completed; and (4) participating in other clinical trials or treatments that may have affected this study. They were randomly assigned to the trial and control groups with 48 participants each. There were no significant differences between the groups regarding baseline characteristics ($P > 0.05$; Table 3).

Effect on quality of life of patients with cancer pain

At the beginning of the study, no significant differences were observed in the scores of each quality of life subscale between the groups ($P > 0.05$; Table 4). Four weeks later, the trial group exhibited higher scores in emotional, cognitive, and social functioning, as well as overall health compared to the control group; conversely, the trial group demonstrated lower scores in pain, constipation, nausea and vomiting, and insomnia compared to the control group. The differences were statistically significant ($P < 0.05$, Table 5).

User satisfaction

The satisfaction levels of patients in the trial group with the app were examined: 35 patients (72.9% of 48 patients) reported being very satisfied, 10 (20.8%) expressed satisfaction, 2 (4.2%) indicated general satisfaction, 1 (2.1%) expressed dissatisfaction, and none reported being "very dissatisfied." Overall, 93.7% of participants reported some level of satisfaction with the app.

Table 2 Evaluation of app module content importance and expert coordination

	Round 1			Round 2		
Modules	Mj	Kj	CV	Mj	Kj	CV
Pain self-management	4.70	0.87	0.14	4.91	0.96	0.08
Online consultation	4.61	0.74	0.16	4.91	0.96	0.08
Reminder	4.61	0.74	0.14	4.87	0.91	0.09
Examination report upload function	4.57	0.65	0.13	4.70	0.83	0.13
Soothing music	4.30	0.43	0.18			
Cancer pain science dissemination	4.65	0.83	0.15	4.83	0.91	0.12

Mj module importance assignment average, *Kj* module full score, *CV* coefficient of variation

Table 3 Patient participants' characteristics

	Control (n = 48)	Trial (n = 48)	P-value
Age (years), mean (SD)	49.85 ± 9.74	52.12 ± 8.95	0.2336
Sex, n (%)			0.5271
Male	16	20	
Female	32	28	
Primary diagnosis, n (%)			0.3109
Esophageal cancer	6	2	
Gastric cancer	6	7	
Intestinal cancer	9	13	
Nasal pharyngeal cancer	2	4	
Lung cancer	14	9	
Pancreatic cancer	2	1	
Breast cancer	2	5	
Ovarian cancer	3	1	
Uterine cancer	1	N/A	
Bone cancer	2	N/A	
Thoracic cancer	1	N/A	
Bladder cancer	N/A	2	
Neck cancer	N/A	1	
Soft tissue sarcoma	N/A	1	
Mediastinal malignancy	N/A	1	
Lymphoma	N/A	1	

N/A not available

Discussion

This study highlights the potential of mobile health technology, particularly tailored apps, in enhancing the management of cancer pain, improving patient outcomes, and addressing gaps in medical resource accessibility. By addressing the deficiencies of existing apps, which lacked a systematic self-management program and interactive features, this study establishes a significant precedent for future research and the formulation of effective interventions in managing cancer pain. Additionally, in the development of a cancer pain self-management app aligned with China's national policies, we took into account the varying levels of medical technology advancement across different provinces; consequently, we chose representative provinces and experts based on the medical progress of each province [20].

The authoritative degree of the experts in this study was greater than 0.70, indicating that the consultation results are likely to be reliable. Further, the findings showed that the expert consultation had high representativeness and credibility. Based on these results, we developed an app containing five modules: pain self-management, online consultation, reminder, examination report upload function, and cancer pain science dissemination. A preliminary study was conducted, which demonstrated that the app improved cancer pain management, with user satisfaction at around 91%. This mobile app integrates effective means of cancer pain self-management, something that is not present in various other similar apps.

Table 4 Baseline quality of life scores by groups and domain based on the European Organization for Research and Treatment of Cancer-Core 30 Questionnaire

Domain	Control (n = 48)	Trial (n = 48)	U	P-value
Function subscales				
Physical functioning	60 (45, 73.33)	53.33 (40, 73.33)	1100.5	0.707
Role functioning	66.67 (33.33, 66.6)	50 (29.17, 66.67)	1077.5	0.577
Cognitive functioning	41.67 (33.33, 58.3)	45.83 (33.33, 58.3)	1173.5	0.876
Emotional functioning	66.67 (66.67, 83.3)	66.67 (62.5, 83.33)	995	0.229
Social functioning	50 (33.33, 66.67)	33.33 (33.33, 66.6)	1022	0.324
Symptom subscales/items				
Dyspnea	0 (0, 33.33)	0 (0, 33.33)	1272	0.3
Insomnia	33.33 (33.33, 66.6)	66.67 (33.33, 66.6)	1227	0.549
Appetite loss	33.33 (33.33, 66.6)	33.33 (33.33, 66.6)	1186	0.781
Nausea/vomiting	16.67 (16.67, 33.3)	33.33 (0, 33.33)	1155	0.985
Constipation	33.33 (0, 41.67)	33.33 (0, 66.67)	1246	0.47
Diarrhea	0 (0,0)	0 (0,0)	1056.5	0.225
Fatigue	55.56 (33.33, 66.6)	55.56 (33.33, 66.6)	1224	0.589
Pain	66.67 (50, 66.67)	66.67 (50, 83.33)	1326.5	0.181
Financial difficulties	66.67 (33.33, 66.6)	66.67 (33.33, 66.6)	1271	0.337
Global health/quality of life	33.33 (33.33, 41.67)	33.33 (25, 41.67)	993	0.23

Table 5 Comparison of quality of life scores after four weeks by groups and domain based on the European Organization for Research and Treatment of Cancer-Core 30 Questionnaire

Domain	Control ($N=48$)	Trial ($N=48$)	U	P
Function subscales				
Physical functioning	56.67 (40, 73.33)	66.67 (36.67, 80)	1252.5	0.462
Role functioning	50 (33.33, 66.67)	66.67 (33.33, 66.6)	1380.5	0.082
Cognitive functioning	45.83 (33.33, 58.3)	83.33 (66.67, 100)	2121	< 0.001
Emotional functioning	66.67 (66.67, 83.3)	83.33 (66.67, 87.5)	1532.5	0.003
Social functioning	33.33 (33.33, 66.6)	66.67 (45.83, 66.6)	1484.5	0.01
Symptom subscales/items				
Dyspnea	0 (0, 33.33)	0 (0, 0)	1108.5	0.677
Insomnia	66.67 (33.33, 66.6)	33.33 (0, 33.33)	517	< 0.001
Appetite loss	33.33 (33.33, 41.6)	33.33 (0, 33.33)	939	0.078
Nausea/vomiting	25 (0, 33.33)	0 (0, 16.67)	679.5	< 0.001
Constipation	33.33 (0, 66.67)	0 (0, 33.33)	671.5	< 0.001
Diarrhea	0 (0, 0)	0 (0, 0)	1151	0.992
Fatigue	55.56 (33.33, 66.67)	33.33 (33.33, 44.4)	681.5	< 0.001
Pain	50 (33.33, 66.67)	16.67 (0, 33.33)	430.5	< 0.001
Financial difficulties	66.67 (33.33, 66.67)	66.67 (33.33, 75)	1345.5	0.129
Global health/quality of life	33.33 (25, 33.33)	50 (50, 58.33)	1988	< 0.001

A full-course medical service model for cancer pain was created based on the app. The app could significantly improve the quality of life of patients with cancer pain in areas such as emotional, cognitive, and social functions ($P < 0.05$), which can be attributed to several factors related to the features and functionality of the self-management app used in the study.

Real-time communication platform

The app provided a platform for real-time communication between patients and healthcare providers, allowing for immediate interaction and support. This feature likely facilitated timely provision of mental and psychological support to patients by qualified professionals, helping them cope with stress, anxiety, and depression.

Disease education and support

The app offered health education resources and guidance on managing cancer pain, including coping strategies for symptoms such as constipation, nausea, and insomnia. By providing patients with information and support to address these challenges, the app may have helped alleviate distress and improve emotional well-being.

Individualized medication guidance

The app likely provided personalized medication guidance to patients, ensuring proper pain management and medication compliance. Studies have shown that adherence to medication regimens can enhance the quality of life for patients with cancer pain by effectively managing symptoms and improving overall well-being.

Increased patient engagement

The app encouraged patients to actively participate in the self-management of their cancer pain by monitoring their condition, tracking symptoms, and engaging with healthcare providers. This increased engagement may have empowered patients to take control of their health, leading to improvements in emotional, cognitive, and social functions.

Supportive care environment

By facilitating communication and support between patients and healthcare providers, the app created a supportive care environment that fostered trust, collaboration, and shared decision-making. This collaborative approach to care likely contributed to the observed improvements in emotional, cognitive, and social functions among patients in the trial group. The multifaceted approach of the self-management app, which combined education, support, communication, and personalized care, likely played a significant role in enhancing emotional, cognitive, and social functions in patients with cancer pain.

According to the treatment principle of NCCN guidelines, opioid drugs are the main type of drug used in the treatment of cancer pain, and symptoms such as nausea, vomiting, and constipation are the main adverse reactions of opioid drugs. Through the app, medical staff regularly provided health education to the patients with cancer pain; guided them on measure to prevent pain; provided timely assistance for them on how to cope with symptoms such as constipation, nausea and vomiting, and insomnia; and guided them to improve sleep and quality of life by relieving their negative emotions. Therefore, app usage was helpful to reduce the occurrence of adverse reactions in patients with cancer pain.

The app helped patients to pay close attention to the development of their own condition and to enhance their enthusiasm to participate in the self-management of cancer pain. The app also provided patients with individualized medication guidance, psychological support, and improved patients' medication compliance. Studies have shown that increased adherence to medication promotes standardized management of cancer pain, improving the quality of life of patients with cancer pain [21].

The study findings also support the hypothesis that a tailored app designed for cancer pain management can be an effective intervention tool for promoting quality of life among patients with cancer pain. The high level of patient satisfaction (93.7%) with the app indicates that the intervention was well-received and positively impacted the participants' experiences and outcomes.

The study's results not only confirm the initial research questions regarding the effectiveness of the self-management app in improving quality of life but also provide empirical evidence to support the hypothesis that a carefully designed and user-friendly app can be a valuable tool in enhancing the well-being of patients with cancer pain.

Conclusions

This study developed a cancer pain self-management app that significantly improved the quality of life for patients compared to standard care. The app included modules for pain management, online consultation, reminders, and education. Patients showed high satisfaction with the app. The findings suggest that tailored apps can effectively enhance patient outcomes in cancer pain management. Practical implications include improved patient engagement, personalized care, enhanced medication compliance, and bridge the gap in medical resource accessibility, especially in underdeveloped rural areas. This intervention tool could serve as a valuable resource for healthcare professionals in optimizing cancer pain management strategies and enhancing patient outcomes. While the results are promising, further research with larger sample sizes and longer follow-up periods is needed to fully understand the effectiveness.

Limitations

The study's sample size of 96 patients may not be representative of the entire population of patients with cancer pain. Future research could involve larger and more diverse samples to enhance the generalizability of the findings. The recruitment of patients during hospital visits may introduce selection bias, as those willing and able to participate may differ from the broader population of patients with cancer pain. Future studies could consider alternative recruitment strategies to reduce bias.

Abbreviations CV: Coefficient of variation; NCCN: National Comprehensive Cancer Network Clinical

Author contribution LZ W and J Y carried out the studies, participated in collecting data, and drafted the manuscript. MB L and WL L performed the statistical analysis and participated in its design. XX L participated in acquisition, analysis, or interpretation of data and draft the manuscript. All authors read and approved the final manuscript.

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Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Ethics approval All procedures were performed in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki and its later amendments. This study was approved by the Medical Ethics Committee of Fujian Medical University Union Hospital (2018KY019). All patients provided written informed consent prior to treatment. The study was carried out in accordance with the applicable guidelines and regulations.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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